
Piecewise-Time-Invariant Neural Granger Causality Detection

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Abstract

We study Granger-causal graph recovery in multivariate time series with time-varying Granger-causality, in particular assuming piecewise time-invariance. We utilize a two-step approach: to estimate the regime boundary and then to recover the causal graphs in each regime through already established time-invariant methods. The main contribution is a normalized local cross-regime prediction score: for each candidate boundary, we compare cross-regime prediction error to same-regime prediction error. Thus, the detector is causality-based rather than distribution-based, as it is based on discrepancy in predictive power rather than in marginal mean or covariance. After estimating the boundary, we split the sequence based on the estimated boundary, and we fit local Granger-causal models: linear VAR and neural autoregressive component-wise MLP. On controlled nonlinear VAR experiments, the boundary score substantially outperforms mean, covariance, mean-plus-covariance, midpoint, and random baselines, achieving much lower median boundary error and higher Success@100 over repeated runs. On a two-regime generalized Lorenz system, the boundary score accurately localizes the regime change. The downstream graph recovery results are more modest: local cMLP improves mean F1 and mean normalized SHD compared with a single global cMLP, suggesting that the split is useful, but neural graph recovery remains sensitive to training variability.

1 Introduction

Many important datasets are naturally represented as multivariate time series. Examples include financial data, where different assets evolve jointly over time, and physiological data, where signals such as EEG channels interact dynamically. In these settings, the goal is often not only to predict future values, but also to understand the directed dependence structure between variables. Correlation is usually too weak for this purpose: it is symmetric, static, and does not distinguish whether the past of one variable helps predict the future of another. Granger causality gives a more structured notion of directed predictive dependence. A variable is Granger-causal for another variable if its past improves prediction of the other variable beyond what can be predicted from the remaining history.

Classical Granger-causal discovery is commonly based on linear vector autoregressive models. Neural Granger-causal methods extend this idea by replacing linear autoregressions with nonlinear predictive models. In particular, component-wise neural predictors can be made interpretable by applying structured sparsity to the input groups corresponding to each source variable. This gives a way to recover a directed Granger-causal graph from a nonlinear time series. Thus, the general logic is: prediction gives a notion of influence, structured sparsity turns this influence into a graph, and the graph summarizes directed predictive dependence.

However, this machinery is most natural when one time-invariant Granger graph explains the whole sequence. This assumption can fail in realistic time series. In EEG, a task change or cognitive-state shift may alter the effective dependencies between channels. In financial data, a market regime

shift may change which assets are predictive of others. In such cases, fitting one global graph can mix multiple mechanisms into a single compromised summary. A natural relaxation is piecewise time-invariant structure: within each regime, a fixed Granger-causal graph is meaningful, but across regimes the graph may change.

This raises a segmentation problem. If the regime boundary were known, then one could apply existing time-invariant Granger-causal methods separately inside each segment. The difficult part is that the boundary is usually not known. In this project, we study the simplest nontrivial version of this problem: a multivariate time series with two regimes and one unknown boundary τ . The goal is first to estimate the boundary $\hat{\tau}$, and then to recover local Granger-causal graphs on the estimated regimes.

A standard way to approach boundary detection is to compare adjacent windows using distributional statistics such as the empirical mean or covariance. This is useful when the boundary creates a visible marginal distribution shift. But a change in Granger-causal structure does not have to appear as a large change in mean or covariance. Two regimes can have similar marginal statistics while having different predictive mechanisms. Therefore, for this problem, the boundary score should target predictive mechanism change directly rather than only distributional change.

We propose a normalized cross-regime prediction score for this purpose. For each candidate boundary t , we train a local model on the left side and a local model on the right side. We then compare same-regime prediction error with cross-regime prediction error. If t is not a true boundary, a model trained on one side should transfer reasonably well to the other side. If t is close to a true boundary, transfer should break: the left model should predict the right side poorly relative to held-out left-side data, and the right model should predict the left side poorly relative to held-out right-side data. The score therefore asks whether crossing the candidate boundary is unusually difficult compared with staying within each side.

After estimating $\hat{\tau}$, we split the sequence into two estimated regimes and apply local Granger-causal discovery methods. We use both a linear VAR/Ridge model and a nonlinear cMLP model. We also fit single global models on the full sequence as baselines. This tests whether the estimated split is actually useful for causal recovery, rather than only producing a plausible boundary score.

The contributions of this project are as follows:

1. We formulate piecewise time-invariant Granger-causal discovery with one unknown boundary as a boundary-localization followed by local graph-recovery problem.
2. We define a normalized candidate-specific cross-regime prediction score that targets changes in predictive mechanism rather than only changes in marginal distribution.
3. We evaluate the boundary score on repeated nonlinear VAR experiments against mean, covariance, mean-plus-covariance, midpoint, and random baselines.
4. We apply the full pipeline to a two-regime generalized Lorenz system, estimating the boundary and recovering local Granger-causal graphs with VAR/Ridge and cMLP.
5. We compare local causal inference against single-regime global baselines, showing that boundary localization improves downstream nonlinear causal recovery in the cMLP setting.

2 Related Work

Change-point detection. Offline change-point detection frames boundary estimation as choosing a cost function and search procedure. This motivates the mean-, covariance-, and mean-plus-covariance baselines used here, but those costs target marginal distribution shifts rather than changes in predictive mechanism.

Deep learning for time series. Deep time-series models commonly use autoregressive prediction to learn temporal structure. Our use of prediction is different: prediction error is used as a diagnostic for regime change, not as the final task.

Neural Granger causality. Tank et al. introduce the cMLP neural Granger model used as the downstream nonlinear learner in this project. Their method assumes one time-invariant graph over the whole sequence; our pipeline first estimates a boundary and then applies such learners locally.

Deep-learning-based Granger tests. Hui et al. also use flexible predictive models for nonlinear Granger inference, but their target is testing Granger causality in a given setting, not localizing when the mechanism changes.

Interventional Granger causality. DiffuGC addresses heterogeneous observational and interventional time series. Our heterogeneity is temporal: the sequence is piecewise time-invariant and the main object is an estimated boundary $\hat{\tau}$.

Decomposition-based nonlinear Granger learning. TDFGnet improves nonlinear graph recovery by decomposing and fusing time-series components. This is complementary to our method: a stronger nonlinear learner could replace cMLP after the boundary is estimated.

3 Method

3.1 Problem setup

Let $X_{1:T} = (x_1, \dots, x_T)$ be a d -dimensional multivariate time series. We assume there is an unknown boundary τ such that the predictive mechanism is time-invariant within each regime, but differs across regimes. The goal is to estimate $\hat{\tau}$ and then infer local Granger-causal graphs on $X_{1:\hat{\tau}}$ and $X_{\hat{\tau}+1:T}$.

All adjacency matrices use the convention $A_{ij} = 1$ to mean that source variable i Granger-causes target variable j .

3.2 Forward-backward residual-asymmetry diagnostic

As a motivating diagnostic, we first consider a forward-backward residual-asymmetry score. We fit a forward autoregressive model on the original sequence and a backward autoregressive model on the time-reversed sequence. The backward residuals are then mapped back to the original time index.

Let $\hat{x}_u^{\text{fwd}}$ denote the forward-model prediction of x_u , and let $\hat{x}_u^{\text{bwd}}$ denote the backward-model prediction of x_u after mapping back to the original time axis. For a candidate boundary t and window size h , define

$$s_{\text{FB}}(t) = \left| \frac{1}{h} \sum_{u=t-h+1}^t \|x_u - \hat{x}_u^{\text{fwd}}\|_2^2 - \frac{1}{h} \sum_{u=t+1}^{t+h} \|x_u - \hat{x}_u^{\text{bwd}}\|_2^2 \right|.$$

The diagnostic boundary estimate is

$$\hat{\tau}_{\text{FB}} = \arg \max_{t \in \mathcal{C}} s_{\text{FB}}(t),$$

where $\mathcal{C}$ is the valid candidate set.

This score captures the intuition that prediction may behave differently on opposite temporal sides of a regime change. However, it is not the final boundary score. Because the forward and backward predictors are trained globally and the score uses raw residual levels, it can peak in regions that are simply hard to predict, rather than at the true regime boundary. The final method therefore replaces this diagnostic with a candidate-specific normalized cross-regime prediction score.

3.3 Normalized candidate-specific cross-regime score

The final method is local and candidate-specific. For each candidate boundary $t \in \mathcal{C}$, we fit a left predictor on data before t and a right predictor on data after t . We then compare cross-regime prediction error against same-regime prediction error.

For each candidate t , define

- $E_{L \rightarrow L}(t)$: error of the left model on held-out left-side data,
- $E_{R \rightarrow R}(t)$: error of the right model on held-out right-side data,
- $E_{L \rightarrow R}(t)$: error of the left model when tested on right-side data,
- $E_{R \rightarrow L}(t)$: error of the right model when tested on left-side data.

Let $\epsilon > 0$ be a small numerical constant used to avoid division by zero. The normalized log-ratio boundary score is

$$s_{\log}(t) = \log \left(\frac{E_{L \rightarrow R}(t) + \epsilon}{E_{L \rightarrow L}(t) + \epsilon} \right) + \log \left(\frac{E_{R \rightarrow L}(t) + \epsilon}{E_{R \rightarrow R}(t) + \epsilon} \right).$$

The estimated boundary is

$$\hat{\tau} = \arg \max_{t \in \mathcal{C}} s_{\log}(t).$$

This normalization is important. A raw prediction error can be large because a local region is noisy or hard to predict. The ratio asks whether crossing the candidate boundary is specifically harder than staying within each side.

3.4 Local and global Granger-causal inference

After estimating $\hat{\tau}$, we split the sequence into two estimated regimes. We then fit local Granger-causal models separately on the two segments. This produces regime-specific causal graphs rather than forcing a single global graph to explain the whole time series. Finally, we also fit a single global Granger-causal model on the full sequence as a comparison baseline.

The downstream causal learners are:

- **VAR/Ridge**: a linear vector autoregressive Granger baseline.
- **cMLP**: a nonlinear component-wise MLP Granger model following Tank et al. For each target j , a separate MLP predicts

$$\hat{x}_{t,j} = f_j(x_{t-p:t-1}).$$

The lagged input is grouped by source variable, so the first-layer weights can be partitioned into source-specific blocks. If $W_{j,i}^{(1)}$ is the first-layer block connecting all lags of source i to the network for target j , the Granger score is

$$G_{ij} = \|W_{j,i}^{(1)}\|_F.$$

We train with squared prediction loss and proximal group-lasso updates, prune small groups, remove self-edges, and threshold the non-self scores to obtain $\hat{A}$.

4 Experiments

4.1 Datasets

Controlled nonlinear VAR. We use a two-regime nonlinear VAR generator for the repeated boundary-localization benchmark. This dataset gives explicit ground-truth graphs and boundaries while allowing marginal moments to be controlled so that boundary detection methods are tested on predictive mechanism change rather than only distribution shift.

Let $x_t \in \mathbb{R}^d$ and let $A^{(r)} \in \{0, 1\}^{d \times d}$ be the regime- r source-target adjacency matrix, where $A_{ij}^{(r)} = 1$ means that source variable i influences target variable j . In regime $r \in \{1, 2\}$, we generate

$$x_{t,j} = \rho x_{t-1,j} + \sum_{\ell=1}^p \sum_{i=1}^d A_{ij}^{(r)} W_{\ell ij}^{(r)} \phi(x_{t-\ell,i}) + \varepsilon_{t,j}, \quad j = 1, \dots, d,$$

where ϕ is a nonlinear activation and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2 I_d)$. The two regimes are simulated separately and then concatenated:

$$X_{1:T} = (X_{1:T_1}^{(1)}, X_{1:T_2}^{(2)}), \quad T = T_1 + T_2,$$

so the true boundary is

$$\tau = T_1.$$

The two nonlinear VAR regimes have $d = 6$ variables with lengths

$$T_1 = 7000, \quad T_2 = 3000, \quad \tau = 7000.$$

We construct the two regimes to differ in their Granger-causal graphs while controlling marginal moments across regimes. This makes the benchmark a direct test of whether a method detects changes in predictive structure rather than relying only on mean or covariance shifts.

Two-regime generalized Lorenz system. Let $x(t) \in \mathbb{R}^d$. In regime $r \in \{1, 2\}$, the latent continuous-time dynamics are, for each variable x_i for $i = 1, 2, \dots, d$:

$$\frac{dx_i(t)}{dt} = -\gamma x_i(t) + F + \sum_{j=1}^d A_{ij}^{(r)} W_{ij}^{(r)} x_j(t) + \sum_{j=1}^d A_{ij}^{(r)} U_{ij}^{(r)} \tanh(x_j(t)) + \sum_{\substack{j, \ell \in \text{Pa}_r(i) \\ j \neq \ell}} C_{ij\ell}^{(r)} x_j(t) x_\ell(t).$$

where

$$\text{Pa}_r(i) = \{j : A_{ij}^{(r)} = 1\}.$$

Here $A_{ij}^{(r)} = 1$ means that variable x_j is allowed to directly enter the dynamics of x_i in regime r .

The two Lorenz regimes use $d = 6$ variables with lengths

$$T_1 = 4000, \quad T_2 = 6000, \quad \tau = 4000.$$

Each regime has a fixed ground-truth graph with two incoming causes per target. The graphs are designed so that, for each target, one incoming cause is shared across regimes and one incoming cause changes. This gives a controlled setting where a single global graph can recover stable edges, but should miss regime-specific edges.

We use step size $\Delta t = 5 \times 10^{-4}$, burn-in 1000, forcing $F = 12$, damping $\gamma = 0.6$, linear scale 0.04, nonlinear scale 0.18, pairwise scale 0.01, and observation noise standard deviation 0.01. The system is integrated using fourth-order Runge–Kutta, the two regimes are concatenated, and each coordinate is standardized.

4.2 Baselines and metrics

Task 1: boundary localization. For boundary localization, the goal is to estimate the unknown regime boundary τ . We compare the proposed normalized cross-regime score against five baselines: mean-difference, covariance-difference, mean-plus-covariance, midpoint, and random valid boundary selection.

For an estimated boundary $\hat{\tau}$ and true boundary τ , we report absolute boundary error

$$\text{Err}_{\text{abs}}(\hat{\tau}, \tau) = |\hat{\tau} - \tau|$$

and normalized absolute boundary error

$$\text{Err}_{\text{norm}}(\hat{\tau}, \tau) = \frac{|\hat{\tau} - \tau|}{T}.$$

Across repeated runs, we also report success rates at fixed tolerances and paired win rates against each baseline.

Task 2: graph recovery. For graph recovery, the goal is to estimate the regime-specific Granger-causal adjacency matrices after splitting the sequence. We evaluate both a linear VAR/Ridge Granger baseline and a nonlinear cMLP Granger model. This gives two comparisons: first, cMLP versus VAR/Ridge as nonlinear versus linear Granger recovery methods; second, local cMLP versus global cMLP to test whether splitting the sequence into estimated regimes improves nonlinear causal recovery compared with fitting one time-invariant model on the full sequence.

All adjacency matrices use the source-target convention: $A_{ij} = 1$ means source variable i Granger-causes target variable j . Self-edges are removed before evaluation. The main edge-level metric is F1:

$$\text{F1} = \frac{2 \langle A, \hat{A} \rangle_F}{\langle A, \mathbf{1} \rangle_F + \langle \hat{A}, \mathbf{1} \rangle_F},$$

where the inner products are taken over non-self directed edges. This is equivalent to the harmonic mean of edge precision and recall, but we report F1 directly.

We also report normalized structural Hamming distance:

$$\text{nSHD}(A, \hat{A}) = \frac{\# \text{additions} + \# \text{deletions} + \# \text{reversals}}{d(d-1)}.$$

A reversal is counted as one structural error rather than as one deletion plus one addition. For the local-versus-global comparison, we report mean F1 and mean nSHD across the two regimes. We do not use graph accuracy as a main metric because the adjacency matrices are sparse, so accuracy can be inflated by correctly predicting absent edges while missing true causal dependencies.

5 Results

5.1 Boundary localization results

In a representative nonlinear VAR run, the normalized cross-regime score recovered the true boundary exactly, while the earlier forward-backward residual-asymmetry diagnostic produced a false peak away from the boundary. The repeated experiment is the main quantitative result.

Method	Mean abs.	Median abs.	Mean norm.	Median norm.	Success@100
Ours	433.65	45.0	0.0361	0.0038	0.82
Mean baseline	3176.31	2858.0	0.2647	0.2382	0.03
Covariance baseline	2965.87	2456.5	0.2472	0.2047	0.07
Mean+covariance baseline	3076.11	2812.0	0.2563	0.2343	0.04
Midpoint baseline	1718.03	1604.0	0.1432	0.1337	0.02
Random baseline	3171.69	2979.5	0.2643	0.2483	0.01

Table 1: Boundary localization summary on the controlled nonlinear VAR benchmark over 100 replicates. Lower absolute and normalized errors are better; higher Success@100 is better.

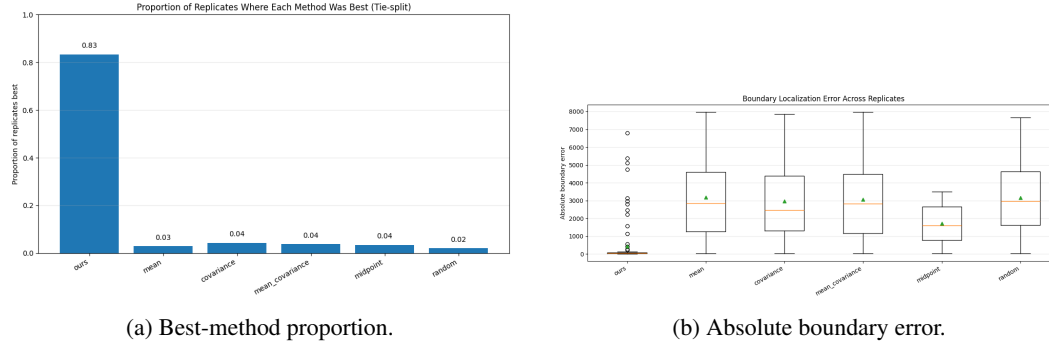


Figure 1: Repeated boundary-localization results. The proposed score is best in most runs and has substantially lower error than the baselines.

5.2 Causal graph recovery results

On the final two-regime Lorenz sequence, the normalized cross-regime score recovered the true boundary exactly: $\tau = 4000$ and $\hat{\tau} = 4000$. We then compared local cMLP, fit separately after the split, with global cMLP, fit once on the full sequence.

Method	Mean F1	Mean nSHD
Global cMLP	0.39 ± 0.09	0.43 ± 0.05
Local cMLP	0.43 ± 0.10	0.40 ± 0.07

Table 2: Repeated cMLP training on the fixed two-regime Lorenz sequence over 10 random initializations. Higher F1 is better; lower nSHD is better.

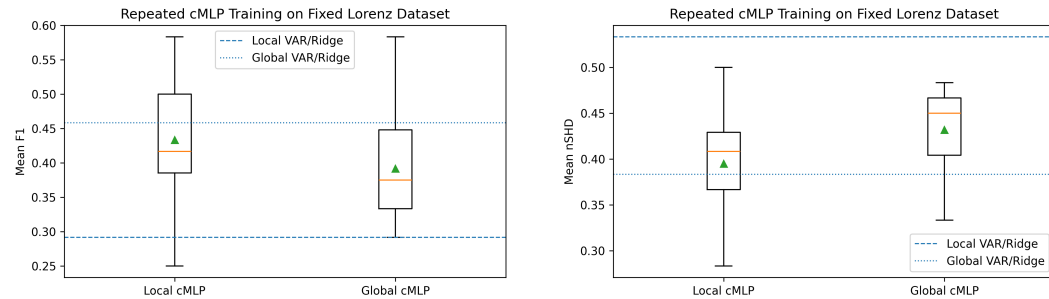


Figure 2: Repeated cMLP training on the fixed two-regime Lorenz sequence. Left: mean F1. Right: mean nSHD.

6 Analysis

The results separate into two questions: boundary localization and downstream graph recovery. For boundary localization, the evidence is strong. The proposed score has mean absolute error 433.65, while the best baseline mean error is 1718.03 from the midpoint baseline. The median error is more decisive: 45.0 for the proposed score, compared with more than 1600 for every baseline. The Success@100 result is also separated: 0.82 for the proposed method, while the strongest baseline reaches only 0.07. Therefore, the method usually localizes the boundary closely, although a few large failures increase the mean error.

The distributional baselines are important because they test the wrong-but-plausible explanation: that the detector is merely finding a mean or covariance shift. Their weak repeated-run performance suggests that the useful signal is not primarily marginal distribution change. The proposed score works because it asks whether predictive transfer across a candidate boundary is worse than same-side prediction.

The graph-recovery result is positive but weaker. On the final Lorenz experiment, the boundary itself is recovered exactly. Local cMLP then improves mean F1 from 0.39 ± 0.09 to 0.43 ± 0.10 and improves mean nSHD from 0.43 ± 0.05 to 0.40 ± 0.07 . Thus, the split helps on average, but the standard deviations and boxplots show substantial overlap. The strongest claim is therefore boundary localization; the downstream local-versus-global result is supportive but not decisive.

7 Discussion

The main limitation is scope. The current setup assumes one boundary and two regimes, while real time series may have several changes, repeated regimes, or gradual transitions. Extending the score to multiple boundaries would require a search procedure such as greedy segmentation, dynamic programming, or a penalized objective.

The experiments are also synthetic. This is appropriate for a controlled first test because the true boundary and graphs are known, and the nonlinear VAR benchmark can suppress easy marginal shifts. However, real EEG or financial data would introduce hidden confounding, nonstationarity, unknown ground truth, and noisier evaluation.

The downstream cMLP remains unstable. Even after a correct split, neural Granger recovery depends on optimization, sparsity strength, lag order, pruning, thresholding, and training duration. A stronger study would include systematic hyperparameter sweeps, more training initializations, stronger change-point baselines, and real-data validation. Finally, the method is causal only in the Granger sense: it detects changes in directed predictive mechanism, not intervention-level causal effects.

8 Conclusion

The boundary score is successful on the controlled nonlinear VAR benchmark. It substantially improves mean error, median error, normalized error, Success@100, and win rate over mean, covariance, mean-plus-covariance, midpoint, and random baselines. It is also meaningfully different from these baselines: it detects failure of predictive transfer rather than marginal distribution shift. Since Granger causality is defined through predictive usefulness, this makes the score causal in the relevant Granger sense.

The split is worth using, but the graph-recovery evidence is more modest. Local cMLP improves mean F1 and mean nSHD over global cMLP, but the improvement is small relative to training variability. Thus, this project supports the premise that segmentation should precede graph recovery when the Granger mechanism changes, but it does not solve the full problem of time-varying neural Granger causality. It solves a restricted two-regime version and identifies normalized predictive transfer as a promising boundary signal.

AI assistance disclosure

I used AI assistance for coding assistance and report organization. I reviewed and verified the code, experiments, and final text.

References

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Appendix

A Additional boundary-localization figures

This appendix collects material removed from the main report to satisfy the page limit. The main report keeps the repeated-run table and compact repeated-run figures. The figures below show the representative nonlinear VAR run and additional boundary-score diagnostics.

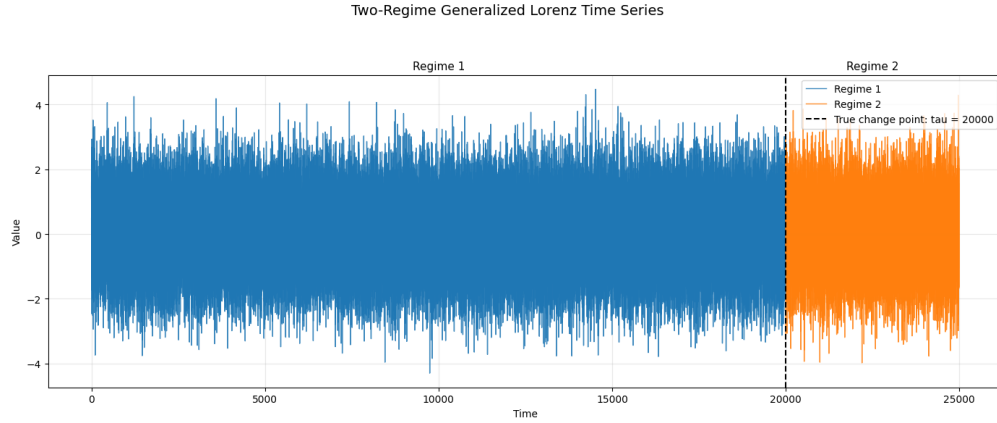
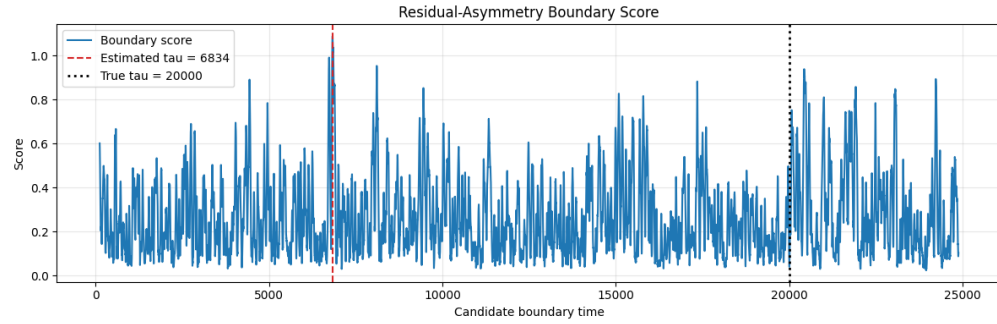
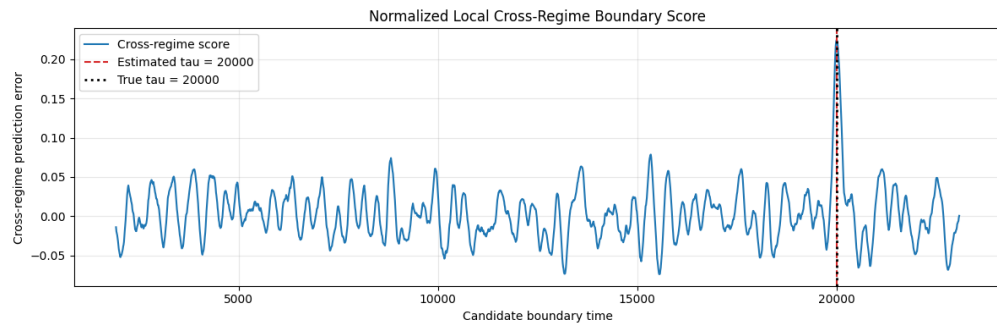


Figure 3: Representative nonlinear VAR synthetic dataset.

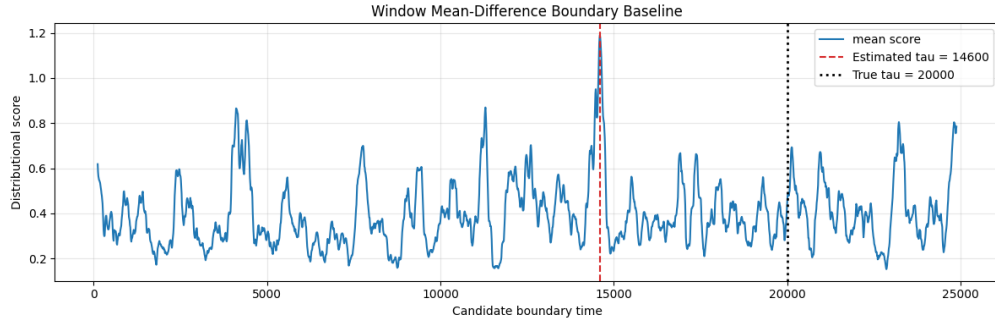


(a) Forward-backward residual-asymmetry diagnostic.

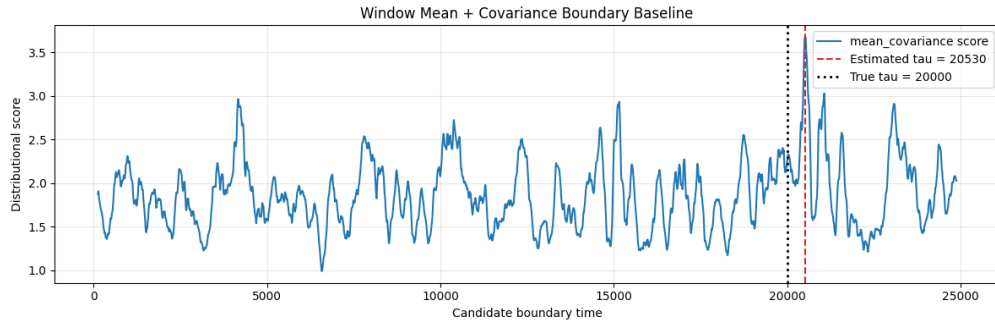


(b) Normalized cross-regime method.

Figure 4: Representative boundary-score comparison. The residual-asymmetry diagnostic can peak away from the true boundary, while the normalized cross-regime score peaks at the true boundary in this run.



(a) Mean baseline.



(b) Mean-plus-covariance baseline.

Figure 5: Additional distributional boundary-detection baselines on representative runs.

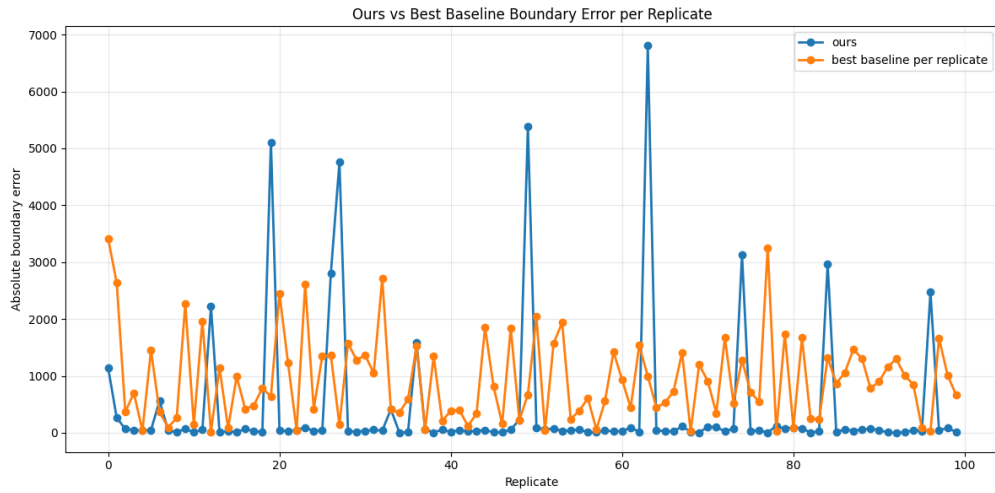


Figure 6: Absolute error curves for the proposed method against the best baseline per replicate.

B Additional Lorenz graph-recovery run

The main report keeps the final repeated Lorenz cMLP experiment. The following earlier Lorenz run is retained here as supplementary material. In this run, the true boundary was $\tau = 3000$ and the estimated boundary was $\hat{\tau} = 2980$, giving absolute error 20 and normalized absolute error 0.002.

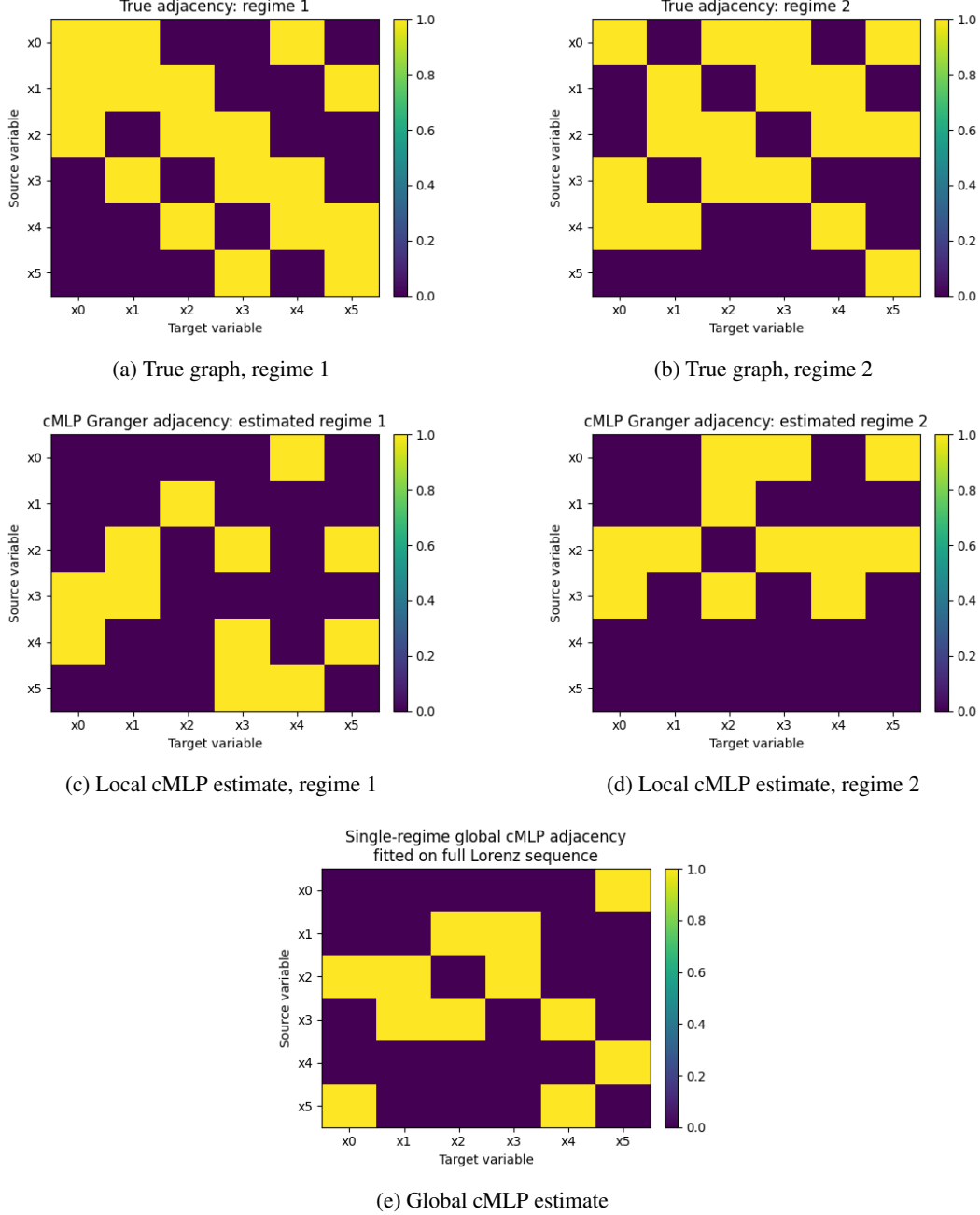


Figure 7: True and estimated Granger-causal graphs on the earlier two-regime Lorenz sequence. The first row shows the true graphs for the two regimes. The second row shows the local cMLP estimates obtained by fitting separate models after boundary localization. The third row shows the global cMLP estimate obtained by fitting a single model to the full sequence.

Learner	Regime	Precision	Recall	F1	nSHD
VAR/Ridge	Regime 1	0.33	0.33	0.33	0.47
VAR/Ridge	Regime 2	0.25	0.25	0.25	0.60
VAR/Ridge	Mean	—	—	0.29	0.53
cMLP	Regime 1	0.50	0.50	0.50	0.37
cMLP	Regime 2	0.67	0.67	0.67	0.27
cMLP	Mean	—	—	0.58	0.32

Table 3: Local Granger-causal graph recovery on an earlier two-regime Lorenz sequence after boundary localization. Higher precision, recall, and F1 are better; lower nSHD is better.

On this earlier Lorenz sequence, the split local cMLP achieved mean F1 about 0.58, while the single global cMLP achieved mean F1 about 0.42. This result was supportive but was moved out of the main report because the final repeated-training comparison is more appropriate for the main text.

C Additional final Lorenz visualization

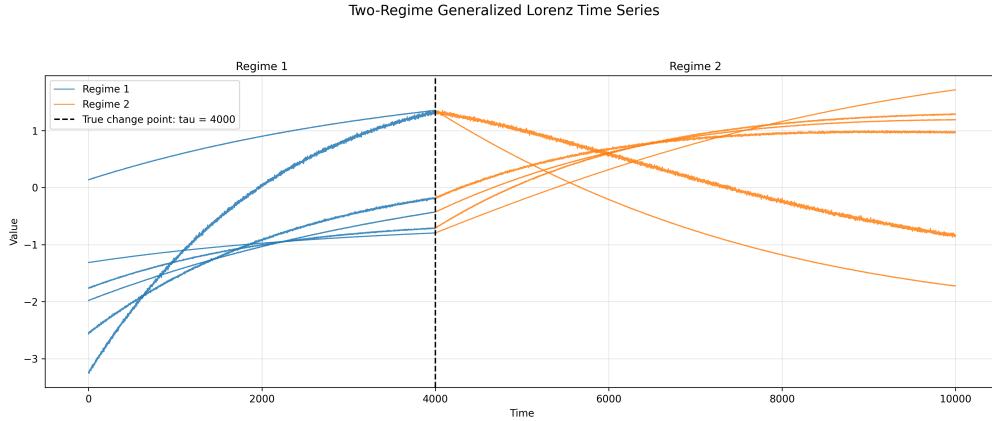


Figure 8: Final two-regime generalized Lorenz six-variate time series. This plot was removed from the main report because the dataset is already described in the experiment section and the boundary-score and graph-recovery results are more important for the page-limited version.

D Expanded analysis, discussion, and conclusion

D.1 Expanded analysis

The results separate into two questions: boundary localization and downstream graph recovery.

Boundary localization. The boundary-localization experiment gives the strongest evidence in the project.

- The proposed normalized cross-regime score has the best mean absolute error: 433.65, compared with 3176.31 for the mean baseline, 2965.87 for the covariance baseline, 3076.11 for the mean-plus-covariance baseline, 1718.03 for the midpoint baseline, and 3171.69 for the random baseline.
- The median error is even more informative. The proposed method has median absolute error 45.0, while all baselines have median errors above 1600. This means that the proposed score usually localizes the boundary very close to the truth, even though it has a few large failures.
- The Success@100 result also supports this interpretation. The proposed method succeeds within 100 time steps in 82% of runs, while the strongest baseline only reaches 7%.
- The win-rate figure is consistent with the table: the proposed method is best in about 83% of replicates. No baseline is best in more than about 4% of replicates.

Therefore, the boundary score is not merely slightly better than the baselines. On this benchmark, it changes the scale of the error. The mean and covariance baselines often fail because the data were constructed so that marginal distribution shifts are not the main signal. The normalized cross-regime

score succeeds because it measures whether predictive transfer across a candidate boundary is worse than same-side prediction.

The representative run also illustrates why the normalized score replaced the forward-backward diagnostic. The forward-backward residual-asymmetry score produced a false peak away from the true boundary, while the normalized cross-regime score peaked at the true boundary. This supports the design choice: raw residual behavior can be large in hard-to-predict regions, while the normalized score asks whether crossing the candidate boundary is unusually difficult relative to staying within each side.

Graph recovery. The graph-recovery results are positive, but weaker than the boundary-localization results. In the final Lorenz run, the true boundary is $\tau = 4000$ and the estimated boundary is $\hat{\tau} = 4000$, so the segmentation step succeeds on this example.

The downstream cMLP comparison asks whether this split helps graph recovery. Across repeated cMLP initializations, the results are

Global cMLP: 0.39 ± 0.09 mean F1, Local cMLP: 0.43 ± 0.10 mean F1,

and

Global cMLP: 0.43 ± 0.05 mean nSHD, Local cMLP: 0.40 ± 0.07 mean nSHD.

Thus, local cMLP is better on both reported graph metrics: higher F1 and lower nSHD. However, the improvement is modest. The standard deviations are large relative to the gap between the means, and the boxplots show overlap between local and global cMLP. Thus, the graph-recovery result supports the usefulness of the split, but it is not as decisive as the boundary-localization result.

D.2 Expanded discussion

The project worked best at the boundary-localization level. The normalized cross-regime score is well matched to the problem because it does not ask whether the left and right windows have different means or covariances. Instead, it asks whether a model trained on one side fails to transfer to the other side. This is closer to the Granger-causal object of interest, since Granger causality is defined through predictive usefulness.

The current setup assumes exactly two regimes and one boundary. This is a useful first case, but it is not the full time-varying problem. Real time series may have several regime changes, repeated regimes, or gradual transitions. A natural next step is to extend the score from a single candidate boundary to a multi-boundary objective. That would require either greedy search, dynamic programming, or a penalized segmentation criterion.

Both main datasets are synthetic. This is not a flaw for a first controlled experiment, because the true boundary and true graphs are known. The nonlinear VAR benchmark is especially useful because it tests whether the method can detect predictive-mechanism change when marginal statistics are controlled. But synthetic success does not prove real-data success. Real EEG or financial data would introduce noise, hidden confounding, nonstationarity, and unknown ground truth.

The cMLP results show that the split helps on average, but the training variability is high. This means that boundary localization is not the only difficulty. Even after a good split, neural Granger recovery depends on optimization, sparsity strength, lag order, pruning, thresholding, and training duration. In this project, cMLP is used mainly as a downstream learner, not as a fully optimized contribution. A stronger version of the project would include a systematic hyperparameter sweep and more repeated training runs.

The boundary baselines are intentionally simple: mean, covariance, mean-plus-covariance, midpoint, and random. These are good sanity checks because they test whether the proposed score is merely detecting distribution shift or guessing a typical location. But they are not the strongest possible change-point baselines. Future work should compare against stronger offline change-point methods, including kernel, likelihood, and predictive-cost methods.

Finally, the score is causal only in the Granger sense. It detects changes in directed predictive mechanism, not intervention-level causality. This distinction matters. The method can support claims about time-varying Granger structure, but it does not establish causal effects under interventions.

D.3 Expanded conclusion

The boundary score is successful. On the repeated nonlinear VAR benchmark, it substantially outperforms all baselines in mean error, median error, normalized error, Success@100, and win rate. The most important number is the median absolute error: 45.0 for the proposed method versus more than 1600 for every baseline. This shows that the score usually finds the boundary very accurately.

The score is also meaningfully different from the baselines. The mean and covariance baselines look for marginal distribution changes. The proposed score instead compares cross-regime prediction error to same-regime prediction error. Therefore, it is not just another distributional change-point score. It targets predictive mechanism change. Since Granger causality is based on predictive improvement, this makes the score Granger-causal in the relevant sense.

For local versus global graph recovery, the split appears worth it, but the evidence is more modest. Local cMLP improves mean F1 from 0.39 to 0.43 and improves mean nSHD from 0.43 to 0.40. This supports the idea that fitting one global graph can blur regime-specific structure. However, the overlap between repeated runs means the result should be read as positive but not decisive.

The problem is worth pursuing. The experiments support the basic premise: if the Granger-causal mechanism changes over time, then segmentation should come before graph recovery. But this project does not solve the full problem of time-varying neural Granger causality. It solves a restricted two-regime version and shows that normalized predictive transfer is a promising boundary signal. A full solution would need multiple-boundary detection, stronger neural graph learners, more stable optimization, stronger baselines, and real-data validation.